Discrete Structures (CS1005)

Mid I Exam (Fall 2026) — Solution

Question 1: Propositional Logic

(a) Logical expressions

i. The dragon is not hiding in the castle, and either the dragon is hiding in the castle or the knight does not carry a magic sword:

$$\neg p \wedge (p \vee \neg q)$$

ii. If the kingdom celebrates victory, then either (the princess opens the secret door and the treasure is protected by a spell) or the dragon is not hiding in the castle:

$$t \rightarrow ((r \wedge s) \vee \neg p)$$

(b) Converse, inverse and contrapositive

"Q unless P" means "if not P, then Q". Here Q = "the hallway is recorded" (r) and P = "the security camera is not working" (¬c). So the statement is ¬(¬c) → r, i.e.

Implication: $c \rightarrow r$ (If the security camera is working, then the hallway is recorded.)

Converse: $r \rightarrow c$ (If the hallway is recorded, then the security camera is working.)

Inverse: $\neg c \rightarrow \neg r$ (If the security camera is not working, then the hallway is not recorded.)

Contrapositive: $\neg r \rightarrow \neg c$ (If the hallway is not recorded, then the security camera is not working.)

(c) Prove [(p ∨ q) ∧ (p → r) ∧ (q → r)] → r is a tautology

Step	Expression	Law used
1	$[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$	Given
2	$[(p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee r)] \rightarrow r$	Implication law (a → b ≡ ¬a ∨ b)
3	$[(p \vee q) \wedge ((\neg p \wedge \neg q) \vee r)] \rightarrow r$	Distributive law (factor out r)
4	$[(p \vee q) \wedge (\neg(p \vee q) \vee r)] \rightarrow r$	De Morgan's law
5	$[((p \vee q) \wedge \neg(p \vee q)) \vee ((p \vee q) \wedge r)] \rightarrow r$	Distributive law
6	$[F \vee ((p \vee q) \wedge r)] \rightarrow r$	Negation law (a ∧ ¬a ≡ F)
7	$[(p \vee q) \wedge r] \rightarrow r$	Identity law (F ∨ a ≡ a)
8	$\neg[(p \vee q) \wedge r] \vee r$	Implication law
9	$\neg(p \vee q) \vee \neg r \vee r$	De Morgan's law
10	$\neg(p \vee q) \vee T$	Negation law (¬r ∨ r ≡ T)
11	T	Domination law

Since the expression simplifies to T, it is a tautology.

(d) Truth values (universe: $\mathbb{R}$)

i. $\forall x \exists y\ (x^2 = y)$ — TRUE. For any real x, choose $y = x^2$, which is a real number.

ii. $\forall x \exists y\ (x^2 - y^2 = 1)$ — FALSE. We need $y^2 = x^2 - 1$. Counterexample: $x = 0$ gives $y^2 = -1$, which has no real solution.

iii. $\exists y \forall x\ (x^2 = y)$ — FALSE. No single y equals every square: $x = 0$ forces $y = 0$, but $x = 1$ forces $y = 1$.

iv. $\exists y \forall x\ (x^2 - y^2 = 1)$ — FALSE. For any y, take $x = 0$: then $-y^2 = 1$, i.e. $y^2 = -1$, impossible in $\mathbb{R}$. So no y works for all x.

(e) Quantified expressions

i. Every prime number greater than 2 is odd:

$$\forall x\,[(P(x) \wedge G(x)) \rightarrow O(x)]$$

ii. All expensive books are good:

$$\forall x\,[(B(x) \wedge E(x)) \rightarrow G(x)]$$

(f) Nested quantifiers

i. Either every customer buys something, or every customer recommends someone:

$$\forall x \exists y\ B(x, y) \vee \forall x \exists y\ R(x, y)$$

ii. Every employee knows someone, and that person helps someone:

$$\forall x \exists y\,[K(x, y) \wedge \exists z\ H(y, z)] \quad \equiv \quad \forall x \exists y \exists z\,[K(x, y) \wedge H(y, z)]$$

(g) Translation into English

i. $\neg\exists y(Q(y) \wedge \forall x\ \neg R(x, y)) \equiv \forall y(Q(y) \rightarrow \exists x\ R(x, y))$

English: There is no qualified student whom nobody recommends; that is, every qualified student is recommended by at least one student.

ii. $\neg\exists y(\forall x \exists z\ T(x, y, z) \vee \exists x \forall z\ U(x, y, z)) \equiv \forall y(\exists x \forall z\ \neg T(x, y, z) \wedge \forall x \exists z\ \neg U(x, y, z))$

English: There is no teacher y such that every teacher asks y about some student, or some teacher teaches y about every student. Equivalently, for every teacher y, there is some teacher who asks y about no student, and no teacher teaches y about all students.

(h) Rules of inference

Let s: the server is running, b: the backup server is running, d: the database is available.

Premises: $s \vee b$, $s \rightarrow d$, $b \rightarrow \neg d$, $\neg d$. Conclusion: $b \wedge \neg d$.

Step	Statement	Reason
1	$s \rightarrow d$	Premise 2
2	$\neg d$	Premise 4
3	$\neg s$	Modus Tollens (1, 2)
4	$s \vee b$	Premise 1
5	b	Disjunctive Syllogism (4, 3)

Step	Statement	Reason
6	$b \wedge \neg d$	Conjunction (5, 2)

Hence the backup server is running and the database is unavailable. (Premise 3 is consistent with this: from b, Modus Ponens on b → ¬d also gives ¬d.)

Question 2: Set Theory

(a)

i. A = {2, 3, 5, 7}, so |A| = 4 and |P(A)| = 2^4 = **16**.

ii. A = {1, 2, 3, 4, 6} (factors of 12 that are ≤ 6) and B = {2, 4, 6, 8, 10}.

Since every element of B is even, x + y can only be an odd prime when x is odd, so only x = 1 and x = 3 can work.

x	Sums x + y	Valid pairs
1	$3, 5, 7, 9, 11$	(1,2), (1,4), (1,6), (1,10)
3	$5, 7, 9, 11, 13$	(3,2), (3,4), (3,8), (3,10)
2, 4, 6	$\text{All even and} \geq 4$	None

Number of ordered pairs = **8**.

(b) Survey of 220 users

Using inclusion–exclusion:

$$|W \cup I \cup F| = 110 + 90 + 80 - 50 - 40 - 30 + 15 = 175$$

i. Users with none = 220 − 175 = **45**.

ii. Exactly one application:

$$\text{Only } W = 110 - 50 - 40 + 15 = 35$$

$$\text{Only } I = 90 - 50 - 30 + 15 = 25$$

$$\text{Only } F = 80 - 40 - 30 + 15 = 25$$

Exactly one = 35 + 25 + 25 = **85**. (Check: exactly two = 35 + 25 + 15 = 75, all three = 15, and 85 + 75 + 15 = 175.)

(c) Set simplification

The complement bars did not come through in the paper's text; the expression is taken in its standard form (A ∩ B) ∪ (A ∩ B′) ∪ (A′ ∩ B) ∪ (A′ ∩ B′).

Step	Expression	Law used
1	$[(A \cap B) \cup (A \cap B')] \cup [(A' \cap B) \cup (A' \cap B')]$	Associative law
2	$[A \cap (B \cup B')] \cup [A' \cap (B \cup B')]$	Distributive law
3	$(A \cap U) \cup (A' \cap U)$	Complement law
4	$A \cup A'$	Identity law
5	U	Complement law

Final set: **U** (the universal set).

(d) Multisets

Multiplicities: A → a:2, b:1, c:3. B → a:1, b:2, c:2, d:1.

i. A ∪ B (maximum multiplicity) = {a, a, b, b, c, c, c, d}

ii. A ∩ B (minimum multiplicity) = {a, b, c, c}

iii. A + B (sum of multiplicities) = {a, a, a, b, b, b, c, c, c, c, c, d}

iv. A − B (multiplicity in A minus in B, if positive) = {a, c}